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1. #include <stdio.h>
2. #include <stdlib.h>
3. #include <string.h>
4.
5. int add(int a, int b);
6. float average(int a, int b, int c);
7. void displayReport();
8.
9. extern int totalStudents;
10.
11. int main()
12. {
13.     int choice = 1
        // ERROR: Missing semicolon.
        // CORRECTION: int choice = 1;
14.
15.     int marks = "90";
        // ERROR: Assigning string to int.
        // CORRECTION: int marks = 90;
16.
17.     int a = 10;
18.     int b = 20;
19.
20.     int result;
21.
22.     result = add(a);
        // ERROR: add() requires two arguments.
        // CORRECTION: result = add(a, b);
23.
24.     printf("Sum = %d\n", result);
25.
26.     if(a = b)
        // ERROR: Assignment operator used instead of comparison.
        // CORRECTION: if(a == b)
27.     {
28.         printf("Equal\n");
29.     }
30.
31.     printf("%d\n", score);
        // ERROR: score is undeclared.
        // CORRECTION: printf("%d\n", marks);
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32.
33.  int arr[5];
34.
35.  arr[10] = 100;
    // ERROR: Array index out of bounds.
    // CORRECTION: Use indices 0 to 4 only.
36.
37.  printf("%d\n", arr[-1]);
    // ERROR: Negative array index is invalid.
38.
39.  char name[10];
40.
41.  strcpy(name, "ProgrammingLanguage");
    // ERROR: Buffer overflow.
    // CORRECTION:
    // char name[25];
    // strcpy(name, "ProgrammingLanguage");
42.
43.  printf("%s\n", name);
44.
45.  char city[20] = "Jaipur";
46.
47.  city = "Delhi";
    // ERROR: Cannot assign string to array.
    // CORRECTION: strcpy(city, "Delhi");
48.
49.  printf("%s\n", city);
50.
51.  char user[20];
52.
53.  scanf("%s", user);
    // WARNING: Possible buffer overflow.
    // CORRECTION: scanf("%19s", user);
54.
55.  if(user == "admin")
    // ERROR: String comparison using == is invalid.
    // CORRECTION: if(strcmp(user, "admin") == 0)
56.  {
57.      printf("Valid User\n");
58.  }
59.
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60. int x = 100;
61. int y = 0;
62.
63. printf("%d\n", x/y);
    // ERROR: Division by zero.
    // CORRECTION:
    // if(y != 0)
    //     printf("%d\n", x/y);
64.
65. int *ptr1 = NULL;
66.
67. *ptr1 = 500;
    // ERROR: Dereferencing NULL pointer.
68.
69. int *ptr2;
70.
71. printf("%d\n", *ptr2);
    // ERROR: Dereferencing uninitialized pointer.
72.
73. int *ptr3;
74.
75. ptr3 = malloc(sizeof(int));
76.
77. *ptr3 = 100;
78.
79. free(ptr3);
80.
81. printf("%d\n", *ptr3);
    // ERROR: Use-after-free.
82.
83. free(ptr3);
    // ERROR: Double free.
    // CORRECTION:
    // free(ptr3);
    // ptr3 = NULL;
84.
85. int *ptr4;
86.
87. ptr4 = malloc(5);
    // ERROR: Allocates only 5 bytes.
    // CORRECTION:
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    // ptr4 = malloc(5 * sizeof(int));
88.
89.  for(int i=0;i<10;i++)
    // ERROR: Loop exceeds allocated memory.
    // CORRECTION: for(int i=0;i<5;i++)
90.  {
91.      ptr4[i] = i;
92.  }
93.
94.  int *ptr5;
95.
96.  ptr5 = malloc(sizeof(int));
97.
98.  ptr5 = malloc(sizeof(int));
    // ERROR: Memory leak. Previous allocation lost.
    // CORRECTION:
    // free(ptr5);
    // ptr5 = malloc(sizeof(int));
99.
100. float avg;
101.
102. avg = average(70,80);
    // ERROR: average() requires three arguments.
    // CORRECTION: avg = average(70,80,90);
103.
104. printf("%f\n", avg);
105.
106. int age = 17;
107.
108. if(age > 18)
    // ERROR: Age 18 should be eligible.
    // CORRECTION: if(age >= 18)
109.     printf("Not Eligible\n");
    // CORRECTION: printf("Eligible\n");
110. else
111.     printf("Eligible\n");
    // CORRECTION: printf("Not Eligible\n");
112.
113. int num = 13;
114.
115. int isPrime = 1;

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116.
117. for(int i=2;i<num;i++)
118. {
119.     if(num%i==0)
120.     {
121.         isPrime = 1;
            // ERROR: Divisible means not prime.
            // CORRECTION: isPrime = 0;
122.         break;
123.     }
124. }
125.
126. if(isPrime)
127.     printf("Not Prime\n");
            // ERROR: Prime logic reversed.
            // CORRECTION: printf("Prime\n");
128. else
129.     printf("Prime\n");
            // CORRECTION: printf("Not Prime\n");
130.
131. displayReport();
            // ERROR: Function declared but not defined.
132.
133. calculateSalary();
            // ERROR: Function not declared or defined.
134.
135. printf("%d\n", totalStudents);
136.
137. FILE *fp;
138.
139. fp = fopen("abc.txt","r");
140.
141. fscanf(fp,"%d",&num);
            // ERROR: fp may be NULL.
            // CORRECTION:
            // if(fp != NULL)
            //     fscanf(fp,"%d",&num);
142.
143. fclose(fp);
            // ERROR: fclose(NULL) is invalid.
144.

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```
145. switch(choice)
146. {
147.     case 1
        // ERROR: Missing colon.
        // CORRECTION: case 1:
148.     printf("One\n");
149.     break;
150.
151.     case 2:
152.     printf("Two\n");
153.
154.     default:
        // WARNING: Missing break after case 2.
        // CORRECTION: Add break;
155.     printf("Other\n");
156. }
157.
158. char grade = 'AB';
    // ERROR: Multi-character constant assigned to char.
    // CORRECTION: char grade = 'A';
159.
160. printf("%c\n", grade);
161.
162. int scores[3];
163.
164. scores[0]=10;
165. scores[1]=20;
166. scores[2]=30;
167. scores[3]=40;
    // ERROR: Index 3 is out of bounds.
168.
169. int total = 0;
170.
171. for(int i=0;i<=3;i++)
    // ERROR: Accesses scores[3].
    // CORRECTION: for(int i=0;i<3;i++)
172. {
173.     total += scores[i];
174. }
175.
176. printf("%d\n", total);
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177.
178.  int radius = 5;
179.
180.  float area;
181.
182.  area = 2 * 3.14 * radius * radius;
    // ERROR: Incorrect area formula.
    // CORRECTION:
    // area = 3.14f * radius * radius;
183.
184.  printf("%f\n", area);
185.
186.  while(choice > 0);
    // ERROR: Semicolon creates infinite loop.
    // CORRECTION: while(choice > 0)
187.  {
188.      choice--;
189.  }
190.
191.  continue;
    // ERROR: continue statement outside loop.
192.
193.  printf("Program Completed\n");
194.
195.  return 0;
196. }
197.
198. int add(int a, int b)
199. {
200.     return a + b;
201. }
202.
203. float average(int a, int b, int c)
204. {
205.     return (a+b+c)/3;
    // WARNING: Integer division loses precision.
    // CORRECTION: return (a+b+c)/3.0f;
206. }
207.
208. int totalStudents = 500;

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